

Project Overview

This project provides an **interactive dashboard** to analyze and visualize Netflix's movies and TV shows dataset. Below are concise, sectioned explanations for each part of the project.

1. Data Loading & Preprocessing

This section explains how the dataset is loaded and prepared for analysis.

1.1. How is the dataset loaded?

Answer: The dataset `titles.csv` is loaded using `pandas.read_csv()`. It contains details like title, type, genres, production countries, release year, ratings, and popularity metrics.

1.2. How are columns like genres and production countries handled?

Answer: Columns with lists stored as strings (e.g., `genres` and `production_countries`) are converted into Python lists using `eval()` for proper analysis.

1.3. How is missing data handled?

Answer: Columns with null values (e.g., `age_certification`, `imdb_score`) are identified using a **missing values heatmap**. Further analysis can handle missing values as needed.

2. Content Type Analysis

This section explains how movies and TV shows are analyzed.

2.1. How is the distribution of movies vs TV shows visualized?

Answer: Count plots using **Seaborn** and pie charts using **Matplotlib** show the proportion of movies and TV shows on Netflix.

2.2. How are the counts measured?

Answer: `df['type'].value_counts()` is used to count the number of Movies and TV Shows.

3. Country-wise Analysis

This section explains the analysis of Netflix content by production countries.

3.1. How are the top contributing countries identified?

Answer: All countries from the `production_countries` column are aggregated, counted using `Counter`, and the top 10 countries are visualized using a **Plotly bar chart**.

3.2. What insights can be drawn?

Answer: The chart shows which countries produce the most Netflix content, highlighting trends in global content production.

4. Genre Popularity Analysis

This section explains how the most popular genres are analyzed.

4.1. How are top genres determined?

Answer: All genres from the genres column are flattened and counted using Counter. The top 10 genres are displayed using a horizontal **Seaborn bar chart**.

4.2. What insights can be drawn?

Answer: The chart highlights the most common genres on Netflix, indicating user preferences and content focus areas.

5. Content Addition Trends

This section explains how Netflix content growth is analyzed over the years.

5.1. How are trends visualized?

Answer: Content addition trends are grouped by release_year and type and visualized using **Plotly line charts**.

5.2. What insights can be drawn?

Answer: The chart shows yearly growth of movies vs TV shows, helping identify periods of rapid content addition.

6. Top Rated Titles

This section explains how high-quality Netflix titles are identified.

6.1. How are top rated titles selected?

Answer: Titles with imdb_score > 8.5 are filtered and sorted in descending order.

6.2. How are they displayed?

Answer: A **Plotly interactive table** shows the top 15 titles, including their type, release year, and IMDb score.

7. Release Heatmap

This section explains how content release patterns are visualized.

7.1. How is the heatmap created?

Answer: Data is grouped by release_year and type, reshaped using .unstack(), and visualized using **Seaborn heatmap**.

7.2. What insights can be drawn?

Answer: The heatmap highlights years with the highest content release and shows trends in movies vs TV shows over time.

8. Tools & Libraries Used

- **Python Libraries:** pandas, numpy, seaborn, matplotlib, plotly, collections, itertools
- **Visualization:** Matplotlib, Seaborn, Plotly

- **Dashboard/Interface:** Streamlit (optional for interactive exploration)
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9. How to Use This Project

1. Load the dataset using `pandas.read_csv()`.
2. Run preprocessing steps to convert strings to lists.
3. Explore different analyses: content type distribution, top countries, genre popularity, content trends, top-rated titles, and release heatmap.
4. Use interactive Plotly charts for dynamic exploration.
5. Deploy with Streamlit for a full interactive dashboard experience.